

Title of the script: Coding with AI

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Keywords: chatgpt, ai, assistant, code, review, video tutorial

Outline:

- Explaining programming concepts
- Adding files to the project workspace
- Getting our code reviewed
- Asking for the explanation of a program

Visual Cue	Narration
Slide 1 Title Slide	Welcome to the Spoken Tutorial on Coding with AI.
Slide 2 Learning Objectives	In this tutorial, we will learn how to use ChatGPT to • Explain programming concepts and • Get our code reviewed
Slide 3 System Requirements To record this tutorial, I am using • Ubuntu Linux OS v22.04 • Firefox web browser v141.0.4 • Visual Studio Code v1.100.3	To record this tutorial, I am using the following setup.
Slide 4 Pre-requisites	To follow this tutorial, learners should have basic knowledge of: • Using internet, and • Programming in C/C++
Slide 5 Code Files The following code files are required to practice this tutorial. 1. strReverse.cpp 2. abstractCls.cpp These files are provided in the Code Files link of this tutorial page.	The following code files are required to practice this tutorial. These files are provided in the Code Files link of this tutorial page.
Slide 6 Significance of Conversational AI	• Tools like ChatGPT can serve as a reliable coding assistant. • It can help with syntax, explain tricky concepts, review your

	code, and suggest optimizations. It is like a knowledgeable colleague you can ask anytime for guidance.
	Suppose I forgot how to use typedef in C programming. Let us go to ChatGPT and ask how to use typedef.
Go to the left Panel and Click on New Project.	First, let us create a New project.
Highlight Projects on the left side of the page Click on Projects Type name: C/C++ Programming Click on Create Project button.	Projects help us organize our work better. Projects keep chats, files, and custom instructions in one place. Click on Projects and enter a name of your choice. I will add the name, C/C++ Programming. Click on Create Project.
Highlight C/C++ Programming under Projects	Observe that the project is created.
Highlight the name C/C++ Programming above Ask Anything	The name of the project appears above the Chat window. It indicates that we are now in the workspace of our project.
Highlight the Chat window in the project workspace	Let us now ask our queries here.
Type: Explain how to use typedef in C programming with some examples. Press Enter.	Type, Explain how to use typedef in C programming with some examples. Press Enter.
Highlight the generated response in the Chat window	ChatGPT first explains what a typedef is, followed by its syntax and usage.

	Now, suppose I have written some code. I want ChatGPT to review it.
<pre>#include <iostream> #include <string> using namespace std; int main() { string str = "Spoken Tutorial"; string rev = ""; for (int i = str.size() - 1; i >= 0; i--) { rev += str[i]; } cout << "Reversed: " << rev << endl; return 0; }</pre>	I have written a C++ code to reverse a string. The code works fine, but I want ChatGPT to suggest improvements. For that, we can directly copy and paste the code in the Chat window. Then give the prompt.
Highlight strReverse.cpp on the Desktop	I have saved the file strReverse.cpp on my Desktop.
Highlight Add files above the Chat window Click on Add files. Highlight the pop-up window Click on that message and upload the file.	Go to your project, C/C++ Programming. Click on Add files. Click on "Add files". A pop-up window appears stating "Add documents, code files, images, and more." Click on that message and upload the file.
Highlight strReverse.cpp under Project files Click X in the right corner of the pop-up window	Wait for the file to get uploaded. Close this window.
Highlight Chat under C/C++ Programming Type: You are a C++ software engineer. I have written a code, strReverse.cpp which reverses a string. Please review it for correctness, readability, and best practices.	type this prompt Press Enter.
Highlight the generated response in the Chat window	Observe that it first confirms the correctness of the program.

	Next, it checks for readability, best practices, and improvements. If needed, we can ask follow-up questions.
	We can ask ChatGPT to explain some code as well.
Open abstractCls.cpp in visual studio code editor and highlight the following part <pre>class Shape { public: virtual double area() const = 0; };</pre>	I have a class-based code that uses an abstract class. I want to understand why an abstract class is being used in this code.
Highlight abstractCls.cpp on the Desktop	I have saved this program as abstractCls.cpp on my system. Let us add this file to the project workspace. Click on the project titled C/C++ Programming.
Click File icon and upload	Click on the Add Files icon and upload abstractCls.cpp file
Type the command in the ChatGPT: You are a C++ software engineer. I have attached a file abstractCls.cpp which has an abstract class. Go through the code and explain why an abstract class is needed in this code.	In the Chat window, type the following prompt. Press Enter and observe the response. bei
Slide 7 Summary In this tutorial, we have learnt how to use ChatGPT to • Explain programming concepts • Get our code reviewed	With this, we come to the end of this tutorial. Let us summarise.
Slide 8 Assignment As an assignment,	As an assignment, please do the following.

• Create a new project and give the C++ code that reverses a string to ChatGPT. • Ask it to write an equivalent Python code. • Ensure that both programs produce the same results.	
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